MILITARY SPECIFICATION SHEET

CONTACTS, ELECTRICAL CONNECTORS, SOCKET, CRIMP REMOVABLE, SHIELDED, (FOR MIL-C-26482 SERIES I CONNECTORS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the contacts described herein shall consist of this document and the latest issue of Specification MIL-C-39029.

INACTIVE FOR NEW DESIGN

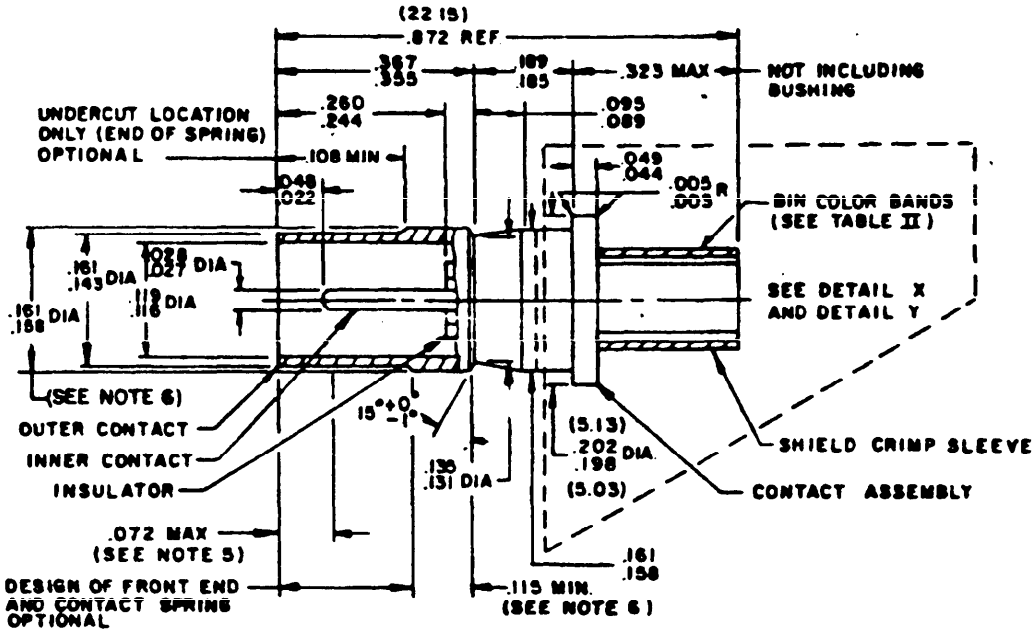
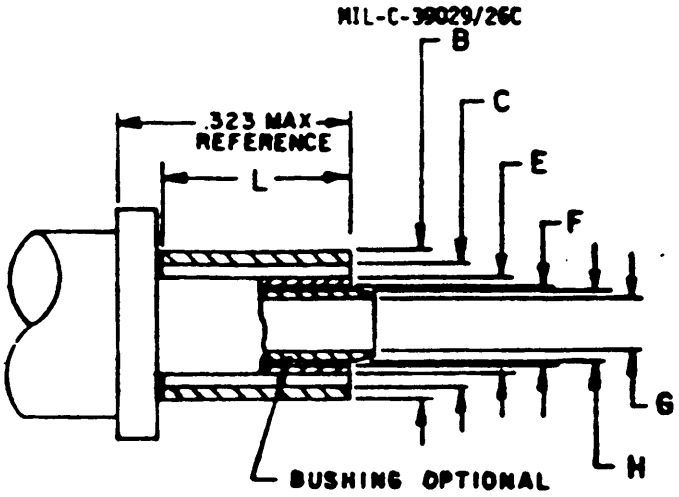


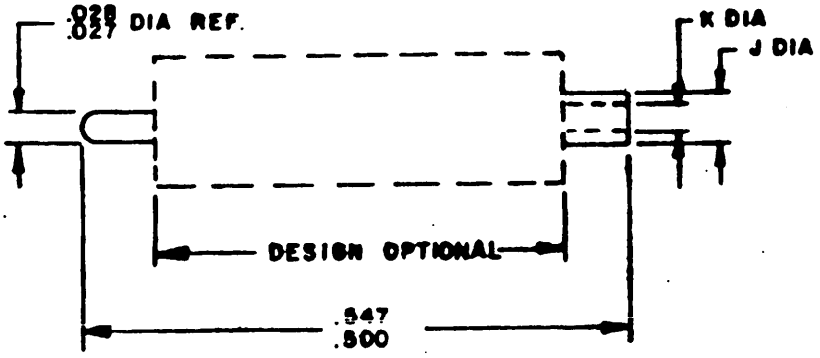
FIGURE 1. Connector contact.

FSC 5935

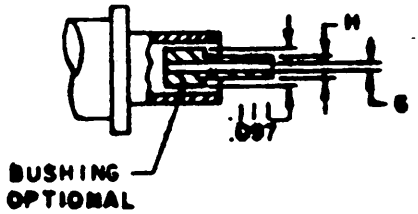
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DETAIL X WIRE TERMINATION END
BIN CODES 207 & 209



INNER CONTACT



DETAIL Y WIRE TERMINATION END
BIN CODE 208

FIGURE 1. Connector contact. -Continued

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BIN code	B	C	E	F	G	H	J	K	L
207	.201 (5.11) .189 (4.80)	.161 (4.09) .151 (3.84)	.136 (3.45) .126 (3.20)	.1150 (2.92) .1065 (2.70)	.069 (1.75) .065 (1.65)	.106 (2.69) .102 (2.59)	.055 (1.40) .045 (1.14)	.030 (.76) .020 (.51)	.270 (6.86) .230 (5.84)
208	---	---	.136 (3.45) .126 (3.20)	.1150 (2.92) .1065 (2.70)	.047 (1.19) .037 (.94)	.076 (1.93) .066 (1.68)	.055 (1.40) .045 (1.14)	.030 (.76) .020 (.51)	.330 (8.38) .290 (7.37)
209	.201 (5.11) .189 (4.80)	.161 (4.09) .151 (3.84)	.136 (3.45) .126 (3.20)	.1150 (2.92) .1065 (2.70)	.058 (1.47) .055 (1.40)	.106 (2.69) .102 (2.59)	.067 (1.70) .057 (1.45)	.044 (1.12) .034 (.86)	.270 (6.86) .230 (5.84)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
3. Metric equivalents are in parentheses for overall length and diameter only.
4. Dimensions shown apply after plating.
5. Indicated dimension to be from front end of contact to point of engagement of socket and a .113 (2.87 mm) diameter square end pin.
6. Indicated diameter applies for indicated length.

FIGURE 1. Connector contact. -Continued

BIN code	A	B	C
207 209	.468	.250	.156
208	.625	.250	.156

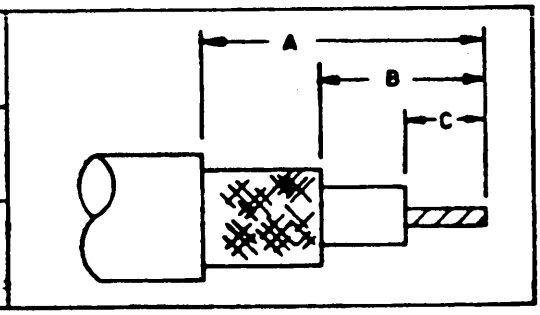
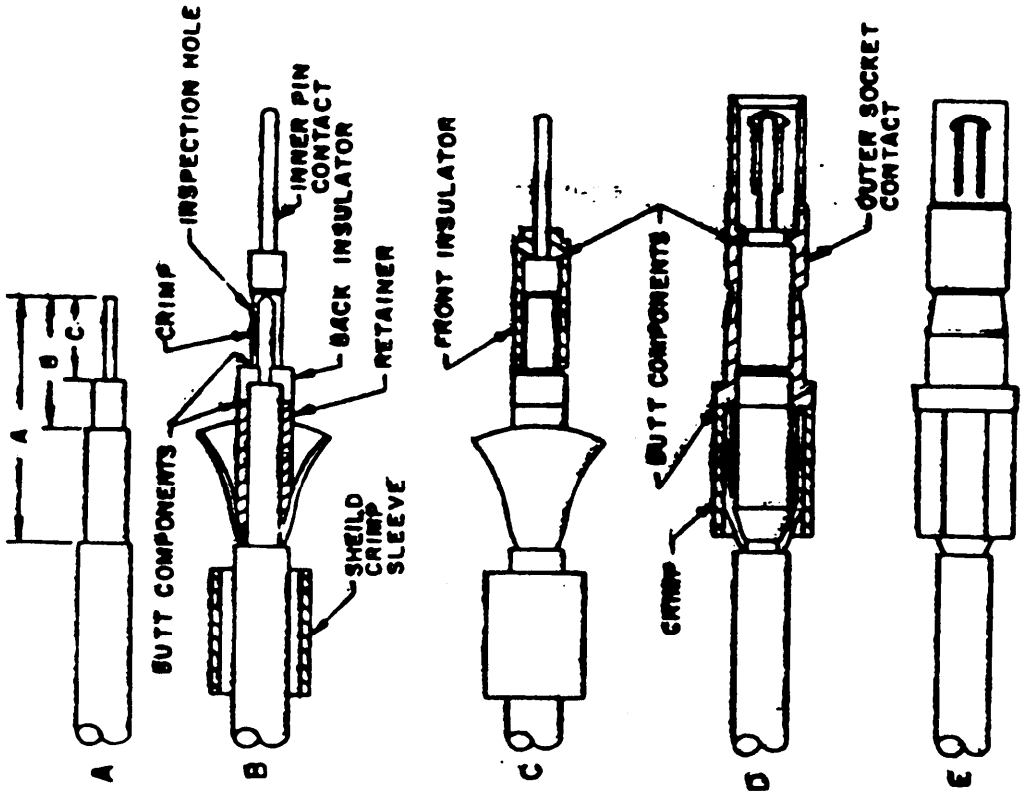


FIGURE 2. Cable stripping dimensions.

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ASSEMBLY INSTRUCTIONS

A Strip cable as illustrated. Ends must be cut cleanly and at right angles to the axial plane of the cable. The cable must not be deformed while making cuts.

B 1. Slide shield crimp sleeve back over cable jacket.
2. Slide retainer over the cable core and under the cable shield. Retainer must bottom against cable shield.
3. Slide back insulator over center conductor and cable core until it butts against the retainer.
4. Slide inner pin contact over cable center conductor. Cable center conductor must be visible through inspection hole in the inner pin contact. Contact must butt against the back insulator.
5. Crimp inner pin contact using crimp tool and positioner listed in table III.

C Slide front insulator, large end first, over the inner pin contact until insulator seats against inner pin contact shoulder.

D 1. Slide outer socket contact over insulator and retainer assembly. Outer socket contact must butt against front insulator.
2. Bring shield crimp sleeve forward over cable shield and butt against outer socket contact shoulder.
3. Crimp shield crimp sleeve using tool and dies listed in table IV.

E Illustrates final assembly.

FIGURE 3. Assembly procedure.

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REQUIREMENTS:

Design and construction:
Dimensions and configuration: See figure 1 and tables I and II.
Tools: See table III.
Assembly procedure: See figure 3. Manufacturer's recommended shipping instructions shall be shipped with the unit package.
Mating contact: See table I.
Cable stripping: See figure 2.
Cable accommodation: See table I.

Low signal level contact resistance (inner contact only): See table IV.

Contact resistance: See table V.

Contact engagement and separation force: The engagement depth shall be as encountered in normal service. The test pins shall be in accordance with MS3197, except the diameters shall be as specified in table VI and surface roughness shall not exceed 3.

Crimp tensile strength (inner and outer contact crimp joint): See table VII.

Vibration: Method 2005 of MIL-STD-1344, test condition IV.

Shock (specified pulse): Method 2004 of MIL-STD-1344, test condition A.

Dielectric withstanding voltage: Applied between inner and outer contact: See table VIII.

Part number: See table IX.

QPL evaluating activity: Naval Avionics Center, Code D-914, Indianapolis, Indiana 46218.

International interest: NEPR 57.

Revision letters are not used to denote changes due to the extensiveness of the changes.

TABLE I. Reference data.

BIN code	Cable data					Mating contact M39029/25-
	Cable accommodated	O.D. of jacket	Number of shields	Dielectric O.D.	Inner conductor size	
207	M17/119-RG174	.100	1	.060	No. 26 AWG 7/.0067	204
	M17/094-RG179	.105	1	.063	No. 30 AWG 7/.004	
	M17/113-RG316	.102	1	.060	No. 26 AWG 7/.0067	
208	M17/093-RG178	.071	1	.034	No. 30 AWG 7/.004	205
209	No. 22 shielded MIL-C-27500	.094 max	1	.048	No. 22 AWG 19/.0063	206

TABLE II. Design characteristics.

Dash No. and BIN code	Color bands			Contact cavity size	Type	Class
	1st	2nd	3rd			
207	Red	Black	Violet	12	D	A
208	Red	Black	Gray	12		
209	Red	Black	White	12		

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TABLE III. Inner and outer contact and shield crimp sleeve tooling.

BIN code	Inner contact tooling		Shield crimp sleeve tooling		Outer contact coding	
	Basic crimping tool	Positioner	Basic crimping tool	Die part no.	Installing tool	Removal tool
207	M22520/2-01	M22520/2-32	M22520/5-01	M22520/5-10	M81969/17-05	M81969/19-09
208	↓	↓	---	---	↓	↓
209	↓	↓	M22520/5-01	M22520/5-10	↓	↓

TABLE IV. Low signal level contact resistance (inner contact only).

BIN code	Cable accommodated	Maximum contact resistance (milliohms)	
		Initial	After conditioning
207	M17/119-RG174	55	66
	M17/094-RG179	120	144
	M17/113-RG316	55	66
208	M17/093-RG178	120	144
209	No. 22 shielded MIL-C-27500	10	12

TABLE V. Contact resistance.

BIN code	Contact	Cable accommodated	Test current (amps)	Maximum voltage drop (millivolts)		
				25° +3°, -0°		150° +3°, -0°
				Initial	After conditioning	
				Max	Max	Max
207	Inner	M17/119-RG174	1	55	66	88
	Outer		12	85	102	136
	Inner	M17/94-RG179	1	120	144	192
	Outer		12	70	84	112
	Inner	M17/113-RG316	1	55	66	88
	Outer		12	75	90	120
208	Inner	M17/093-RG178	1	120	144	192
	Outer		12	110	132	176
209	Inner	No. 22 shielded MIL-C-27500				
	Outer		12	80	96	96

TABLE VI. Contact engagement and separation force.

Test pin diameter (inches)	Socket contact pair					
	Initial			After conditioning		
	Minimum separation force	Maximum average engagement	Maximum engagement force	Minimum separation force	Maximum average engagement	Maximum engagement force
	oz	oz	oz	oz	oz	oz
.1140 +.0002 -.0000	N/A	N/A	30	N/A	N/A	36
.1120 +.0000 -.0002	3	N/A	N/A	2	N/A	N/A

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TABLE VII. Crimp tensile strength (inner and outer contact crimp joint.

BIN code	Cable accommodated	Axial load (pounds minimum)	
		Inner contact	Outer contact
207	M17/119-RG174 M17/094-RG179 M17/113-RG316	15 3.5 15	15 15 15
208	M17/093-RG178	3.5	10
209	No. 22 shielded MIL-C-27500	15	15

TABLE VIII. Dielectric withstanding voltage (applied between inner and outer contact)

Altitude	Test voltages, ac rms
Sea level 50,000 feet	1000 250

TABLE IX. Part number.

BIN code	Part number	Supersedes
207	M39029/26-207	M39029/26-01
208	M39029/26-208	M39029/26-02
209	M39029/26-209	M39029/26-03

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - AR, MI
Navy - EC, SH, MC
Air Force - 11, 17, 99
DLA - ES

User activities:
Army -
Navy - OS
Air Force -

Preparing activity:
Navy - AS

Agent:
DLA - ES

(Project 5935-3098-10)

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DOCUMENT IDENTIFIER (Number) AND TITLE MIL-C-39029/26C Contact, Electrical Connector Socket, Crimp Removable, Shielded

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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B. RECOMMENDED WORDING CHANGE

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